

Robust Edge Detection for Robot Vision

Image Processing Project

Data and Compute Intensive Sciences Research Group

Balázs, Paszkál, Vince, Levente, Bence, Antal
Éva, Hajni

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Outline

Motivation

The Problem

Our Method

Results

Summer Camp Project

Why Do Machines Need Vision?

Modern robots need to understand their environment.

- ▶ Autonomous vehicles
- ▶ Warehouse robots
- ▶ Drones
- ▶ Industrial systems

A camera only provides pixels.

How can we extract useful information from images?

From Image to Information



- ▶ Raw image contains millions of values
- ▶ Important structures are hidden
- ▶ We search for:
 - ▶ edges
 - ▶ shapes
 - ▶ movement

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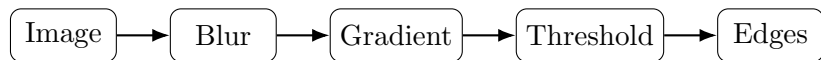
Our Method

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Classical Edge Detection

One common method is the Canny edge detector.



The Limitation

Classical methods can be unstable.



Too many edges



Missing edges

Our Goal

We want a method that:

- ▶ Detects important structures
- ▶ Ignores noise
- ▶ Works with real camera input
- ▶ Is robust in changing conditions

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Convolution Based Approach

Images can be processed using convolution kernels.

$$\begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

The kernel scans the image and detects patterns.

How Convolution Works



The result highlights important information.

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Example Results



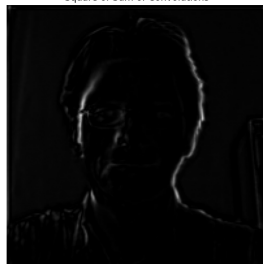
Original

Edges (Canny)



Canny

Square of Sum of Convolutions



Our method

Applications

Applications:

- ▶ Robot vision
- ▶ Video stabilization
- ▶ Distance measurement
- ▶ Object tracking
- ▶ Autonomous systems

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What We Do During The Camp

1. Introduction to image processing
2. Understanding convolution methods
3. Implementing edge detection
4. Applying methods to robot camera
5. Video stabilization experiments
6. Evaluation of results

Final Goal

Build a system that can:

See $\rightarrow$ Understand $\rightarrow$ React

using real camera data.